RPractical_Evangelista

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PART A

A1

```
v <- seq(c(5, 10, 15, 20, 25, 30, 35, 40, 45, 50))
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 55
```

```
mean(v)
```

```
## [1] 5.5
```

A2

```
w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
17 %% 5
```

```
## [1] 2
```

```
17 %/% 5
```

```
## [1] 3
```

```
2^10
```

```
## [1] 1024
```

```
sqrt(144)
```

```
## [1] 12
```

```
round(pi, 3)
```

```
## [1] 3.142
```

A4

```r
a <- c(2,4,6,8)
b <- c(1,3,5,7)
a+b
```

```
## [1]  3  7 11 15
```

```r
a*b
```

```
## [1]  2 12 30 56
```

```r
a>b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a*b)
```

```
## [1] 100
```

A5

```r
paste("Student", 1:3, sep = "-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
class(42)
```

```
## [1] "numeric"
```

```r
class(42L)
```

```
## [1] "integer"
```

```r
class("42")
```

```
## [1] "character"
```

```r
class(41i)
```

```
## [1] "complex"
```

```r
class(TRUE)
```

```
## [1] "logical"
```

PART B

B1

```r
students <- data.frame(
  id = 1:8,
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section = c("A","B","A","C","B","C","A","B"),
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringAsFactors = FALSE
  )
students
```

```
##   id  name sex section midterm final attendance stringAsFactors
## 1  1 Alice   F       A      78    82         95           FALSE
## 2  2 Bruno   M       B      85    79         88           FALSE
## 3  3  Chen   M       A      92    96        100           FALSE
```

```
## 4  4  Dana   F       C      67    71         72           FALSE
## 5  5  Elif   F       B      74    80         90           FALSE
## 6  6 Farid   M       C      88    91         85           FALSE
## 7  7 Grace   F       A      95    93         98           FALSE
## 8  8  Hiro   M       B      59    65         60           FALSE
```

B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  8 variables:
##  $ id             : int  1 2 3 4 5 6 7 8
##  $ name           : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex            : chr  "F" "M" "M" "F" ...
##  $ section        : chr  "A" "B" "A" "C" ...
##  $ midterm        : num  78 85 92 67 74 88 95 59
##  $ final          : num  82 79 96 71 80 91 93 65
##  $ attendance     : num  95 88 100 72 90 85 98 60
##  $ stringAsFactors: logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
```

```
dim(students)
```

```
## [1] 8 8
```

```
names(students)
```

```
## [1] "id"              "name"            "sex"             "section"
## [5] "midterm"         "final"           "attendance"      "stringAsFactors"
```

```
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final          attendance     stringAsFactors
##  Min.   :65.00   Min.   : 60.00   Mode :logical
##  1st Qu.:77.00   1st Qu.: 81.75   FALSE:8
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students$average <- round((students$midterm + students$final) / 2,1)
students
```

```
##   id  name sex section midterm final attendance stringAsFactors average
## 1  1 Alice   F       A      78    82         95           FALSE    80.0
## 2  2 Bruno   M       B      85    79         88           FALSE    82.0
## 3  3  Chen   M       A      92    96        100           FALSE    94.0
## 4  4  Dana   F       C      67    71         72           FALSE    69.0
## 5  5  Elif   F       B      74    80         90           FALSE    77.0
## 6  6 Farid   M       C      88    91         85           FALSE    89.5
## 7  7 Grace   F       A      95    93         98           FALSE    94.0
```

```
## 8  8  Hiro   M       B      59    65         60           FALSE    62.0
```

B4

```
students$passed <- students$average >= 70 & students$attendance >= 75
students
```

```
##   id  name sex section midterm final attendance stringAsFactors average passed
## 1  1 Alice   F       A      78    82         95           FALSE    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88           FALSE    82.0   TRUE
## 3  3  Chen   M       A      92    96        100           FALSE    94.0   TRUE
## 4  4  Dana   F       C      67    71         72           FALSE    69.0  FALSE
## 5  5  Elif   F       B      74    80         90           FALSE    77.0   TRUE
## 6  6 Farid   M       C      88    91         85           FALSE    89.5   TRUE
## 7  7 Grace   F       A      95    93         98           FALSE    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60           FALSE    62.0  FALSE
```

B5

```
students$standing <- ifelse(students$average >= 90, "Honors", ifelse(students$average >= 75, "Good", "N
students
```

```
##   id  name sex section midterm final attendance stringAsFactors average passed
## 1  1 Alice   F       A      78    82         95           FALSE    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88           FALSE    82.0   TRUE
## 3  3  Chen   M       A      92    96        100           FALSE    94.0   TRUE
## 4  4  Dana   F       C      67    71         72           FALSE    69.0  FALSE
## 5  5  Elif   F       B      74    80         90           FALSE    77.0   TRUE
## 6  6 Farid   M       C      88    91         85           FALSE    89.5   TRUE
## 7  7 Grace   F       A      95    93         98           FALSE    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60           FALSE    62.0  FALSE
##       standing
## 1         Good
## 2         Good
## 3       Honors
## 4 Need Support
## 5         Good
## 6         Good
## 7       Honors
## 8 Need Support
```

B6

```
classify <- function(score) {
  if (score>=85) {
    "Distinction"
  } else if (score>=70) {
    "Merit"
  } else if (score>=50) {
    "Pass"
  } else {
    "Fail"
  }
}

result <- sapply(c(90,73,55,40), classify)
result
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

PART C

C1

```
new_students <- data.frame(
  id = c(9,10),
  name = c("Ines","Jamal"),
  sex = c("F","M"),
  section = c("C", "A"),
  midterm = c(81,70),
  final = c(84,68),
  attendance = c(92,80),
  stringAsFactors = FALSE
)

new_students$average <- round((new_students$midterm + new_students$final) / 2,1)
new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
                         ifelse(new_students$average >= 75, "Good", "Need Support"))

students_all <- rbind(students, new_students)
students_all
```

```
##    id  name sex section midterm final attendance stringAsFactors average passed
## 1   1 Alice   F       A      78    82         95           FALSE    80.0   TRUE
## 2   2 Bruno   M       B      85    79         88           FALSE    82.0   TRUE
## 3   3  Chen   M       A      92    96        100           FALSE    94.0   TRUE
## 4   4  Dana   F       C      67    71         72           FALSE    69.0  FALSE
## 5   5  Elif   F       B      74    80         90           FALSE    77.0   TRUE
## 6   6 Farid   M       C      88    91         85           FALSE    89.5   TRUE
## 7   7 Grace   F       A      95    93         98           FALSE    94.0   TRUE
## 8   8  Hiro   M       B      59    65         60           FALSE    62.0  FALSE
## 9   9  Ines   F       C      81    84         92           FALSE    82.5   TRUE
## 10 10 Jamal   M       A      70    68         80           FALSE    69.0  FALSE
##        standing
## 1          Good
## 2          Good
## 3        Honors
## 4  Need Support
## 5          Good
## 6          Good
## 7        Honors
## 8  Need Support
## 9          Good
## 10 Need Support
```

C2

```
quiz <- c(8,7,10,6,7,9,10,5,8,6)
students_all <- cbind(students_all, quiz)

print(students_all)
```

```
##    id  name sex section midterm final attendance stringAsFactors average passed
## 1   1 Alice   F       A      78    82         95           FALSE    80.0   TRUE
```

```
## 2   2 Bruno   M       B      85    79         88           FALSE    82.0   TRUE
## 3   3  Chen   M       A      92    96        100           FALSE    94.0   TRUE
## 4   4  Dana   F       C      67    71         72           FALSE    69.0  FALSE
## 5   5  Elif   F       B      74    80         90           FALSE    77.0   TRUE
## 6   6 Farid   M       C      88    91         85           FALSE    89.5   TRUE
## 7   7 Grace   F       A      95    93         98           FALSE    94.0   TRUE
## 8   8  Hiro   M       B      59    65         60           FALSE    62.0  FALSE
## 9   9  Ines   F       C      81    84         92           FALSE    82.5   TRUE
## 10 10 Jamal   M       A      70    68         80           FALSE    69.0  FALSE
##         standing quiz
## 1           Good    8
## 2           Good    7
## 3         Honors   10
## 4  Need Support    6
## 5           Good    7
## 6           Good    9
## 7         Honors   10
## 8  Need Support    5
## 9           Good    8
## 10 Need Support    6
```

```
class(students_all)
```

```
## [1] "data.frame"
```

C3

```
m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
m
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

C4

```
bad <- students_all
names(bad)[1] <- "wrong_name"

tryCatch(rbind(students, bad),
         error = function(e) {
           message("rbind failed: ", conditionMessage(e))
           NULL
         })
```

```
## rbind failed: numbers of columns of arguments do not match
```

```
## NULL
```

```
#If two data frames have different column names, rbind() fails with an error. tryCatch() catches the er
```

PART D

D1

```
print(students_all$name)
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
print(students_all[["name"]])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
print(students_all[, "name"])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

D2

```
print(students_all[2:5, c("name", "midterm", "final")])
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

D3

```
print(students_all[students_all$attendance > 90, ])
```

```
##   id  name sex section midterm final attendance stringAsFactors average passed
## 1  1 Alice   F       A      78    82         95           FALSE    80.0   TRUE
## 3  3  Chen   M       A      92    96        100           FALSE    94.0   TRUE
## 7  7 Grace   F       A      95    93         98           FALSE    94.0   TRUE
## 9  9  Ines   F       C      81    84         92           FALSE    82.5   TRUE
##   standing quiz
## 1     Good    8
## 3   Honors   10
## 7   Honors   10
## 9     Good    8
```

D4

```
res <- subset(students_all,
              sex == "F" & section %in% c("A", "C") & final >= 75,
              select = c(name, section, final))
res
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

D5

```
students_all$name[which.max(students_all$average)]
```

```
## [1] "Chen"
```

```
students_all$name[which.min(students_all$average)]
```

```
## [1] "Hiro"
```

D6

```
sorted <- students_all[order(students_all$average, decreasing = TRUE), ]
print(head(sorted, 3))
```

```
##   id  name sex section midterm final attendance stringAsFactors average passed
## 3  3  Chen   M       A      92    96        100           FALSE    94.0   TRUE
## 7  7 Grace   F       A      95    93         98           FALSE    94.0   TRUE
## 6  6 Farid   M       C      88    91         85           FALSE    89.5   TRUE
##   standing quiz
## 3   Honors   10
## 7   Honors   10
## 6     Good    9
```

PART E

E1

```
students_all$section <- factor(students_all$section)
students_all$sex     <- factor(students_all$sex)

print(levels(students_all$section))
```

```
## [1] "A" "B" "C"
```

```
print(levels(students_all$sex))
```

```
## [1] "F" "M"
```

E2

```
print(table(students_all$sex))
```

```
##
## F M
## 5 5
```

```
print(table(students_all$section))
```

```
##
## A B C
## 4 3 3
```

```
print(table(students_all$sex, students_all$section))
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

E3

```
print(as.integer(students_all$sex))
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

E4

```
students_all$sex_label <- factor(students_all$sex,
                                 levels = c("F", "M"),
                                 labels = c("Female", "Male"))
print(levels(students_all$sex_label))
```

```
## [1] "Female" "Male"
```

E5

```
print(summary(students_all$sex_label))
```

```
## Female   Male
##      5      5
```

```
print(mean(students_all$average[students_all$sex_label == "Female"]))
```

```
## [1] 80.5
```

```
print(mean(students_all$average[students_all$sex_label == "Male"]))
```

```
## [1] 79.3
```